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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

II Semester B. E. Examinations Oct-2023

(Common to AS, CH, IM, ME)

VECTOR CALCULUS AND COMPUTATIONAL METHODS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.
4. Use of mathematics Handbook is permitted. Do not write anything on handbook.

PART-A

1	1.1	If $\phi = 3x^2y - y^3z^2$, then $grad\ \phi$ at the point $(1,-2,-1)$ is _____.	02
	1.2	The value of λ so that the vector $(x+3y)\hat{\imath} + (y-2z)\hat{\jmath} + (x+\lambda z)\hat{k}$ is a solenoidal vector is _____.	02
	1.3	Evaluate $\Delta^2(x^2 - 6x + 2)$ with $h = 1$.	02
	1.4	Find the work done in moving a particle in the force field $\vec{F} = (x-y)\hat{\imath} + (x-2y)\hat{\jmath}$ along the curve $x = 2\cos t, y = 2\sin t$ and $t = 0$ to $t = \pi$.	02
	1.5	If $\vec{F} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and V is the region bounded by the cube $x = 0,\ x = 1, y = 0, y = 1, z = 0, z = 1$, then $\iiint_V \nabla.\vec{F}\ dv$ =_____.	02
	1.6	The equation $z_{xx} + 2xy(z_x)^2 + z_y = 5$ is of order _____ and degree _____.	02
	1.7	By eliminating the arbitrary constants from $z = a^2x + ay^2 + b$, the partial differential equation formed is _____.	02
	1.8	The first approximation to a real root of the equation $x^3 = 5x - 3$ between 0 and 1 by method of false position is _____.	02
	1.9	The first approximate positive root of $3x = \cos x + 1$ by taking $x_0 = 0.5$ using Newton-Raphson method is _____.	02
	1.10	If $y(1) = 0, y(3) = 12, y(4) = 19$, find interpolating polynomial using Lagrange's formula.	02

PART-B

2	a	Find the directional derivative of $f(x,y,z) = x^2y^2z^2$ at the point $(1,1,-1)$ in the direction of the tangent to the curve $x = e^t,\ y = 1 + 2\sin t, z = t - \cos t$ at $t = 0$.	06
	b	Find $div\ \vec{F}$ and $curl\ \vec{F}$ where $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$.	04
	c	Show that $\vec{F} = (2xy^2 + yz)\hat{\imath} + (2x^2y + xz + 2yz^2)\hat{\jmath} + (2y^2z + xy)\hat{k}$ is a conservative force field. Find its scalar potential such that $\vec{f} = \nabla\phi$.	06
3	a	Verify Green's theorem for $\oint_C (xy + y^2)dx + x^2dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.	08

	b	Using divergence theorem, evaluate $\iint_S \vec{A}.\hat{n}\, ds$ where $\vec{A} = x^3\hat{\imath} + y^3\hat{\jmath} + z^3\hat{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$.	08
		OR	
4	a	Verify the Stoke's theorem for the vector $\vec{F} = (x^2 + y^2)\hat{\imath} - 2xy\,\hat{\jmath}$ taken round the rectangle bounded by $x = 0, x = a, y = 0, y = b$.	10
	b	If $\vec{F} = xy\hat{\imath} + yz\,\hat{\jmath} + zx\hat{k}$, evaluate $\int_C \vec{F}.d\vec{r}$ where C is the curve represented by $x = t,\ y = t^2, z = t^3, -1 \le t \le 1$.	06
5	a	Form a PDE by eliminating the arbitrary function f from the relation $z = e^{ax+by}\, f(ax - by)$.	04
	b	Solve $(y - z)p + (z - x)q = (x - y)$.	04
	c	Solve by method of separation of variables, $u_x = 2u_t + u$ when $u(x, 0) = 6e^{-3x}$.	08
		OR	
6	a	Solve $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$.	06
	b	Obtain the solution of wave equation in one dimension by the method of separation of variables.	10
7	a	Find the real root of the equation $x \log_{10} x - 1.2 = 0$ by using Regula-Falsi method. Perform three iterations.	08
	b	Solve $y' = x + y, y(0) = 1$ by Taylor's series method for the values of y at $x = 0.1$ and $x = 0.2$, taking fourth degree terms.	08
		OR	
8	a	Find the positive root of $x^4 - x = 10$ correct to four decimal places, using the Newton-Raphson method.	06
	b	Apply the Runge-Kutta method to find the approximate value of y for $x = 0.2$ in steps of 0.1 if $\frac{dy}{dx} = x + y^2, y = 1$ when $x = 0$.	10
9	a	The table gives the distance in nautical miles of the visible horizon for the given heights in feet above the earth's surface:	

x =height	100	150	200	250	300	350	400
y =distance	10.63	13.03	15.04	16.81	18.42	19.90	21.27

Find the values of y when

i) $x = 160\ ft$ ii) $x = 410\ ft$. 08

	b	Use Lagrange's interpolation formula to find the value of y when $x = 10$ if the following values of x and y are given:			

x	5	6	9	11
y	12	13	14	16

08

OR

10	a	Evaluate $\int_0^6 \frac{dx}{1+x^2}$ using i) Simpson's $\frac{1}{3}rd$ rule ii) Simpson's $\frac{3}{8}th$ rule iii) Weddle's rule by taking 6 sub intervals.	08
	b	Find the cubic polynomial using appropriate interpolation formula which takes the following values.	

x	0	1	2	5
$f(x)$	2	3	12	147

Hence evaluate $f(3)$. 08